

Data Structure(CSL202)

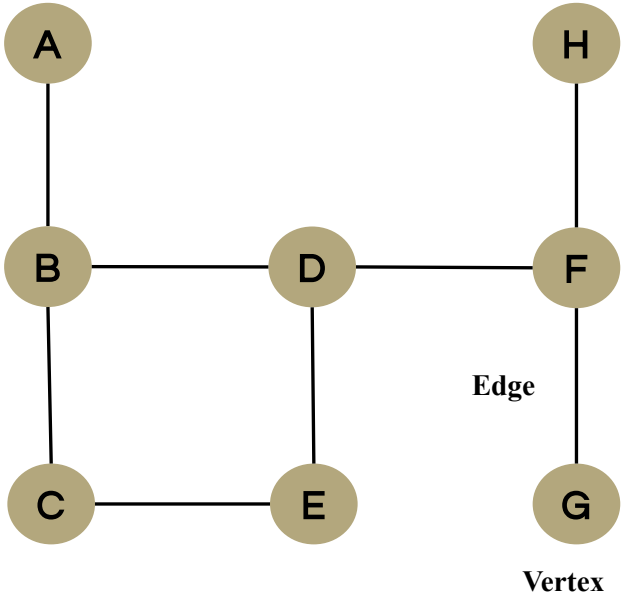
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भारतीय प्रौद्योगिकी संस्थान भिलाई
Indian Institute of Technology Bhilai

Graph

Graph

- A non-linear data structure.
- Collection of vertices (V) and edges (E).

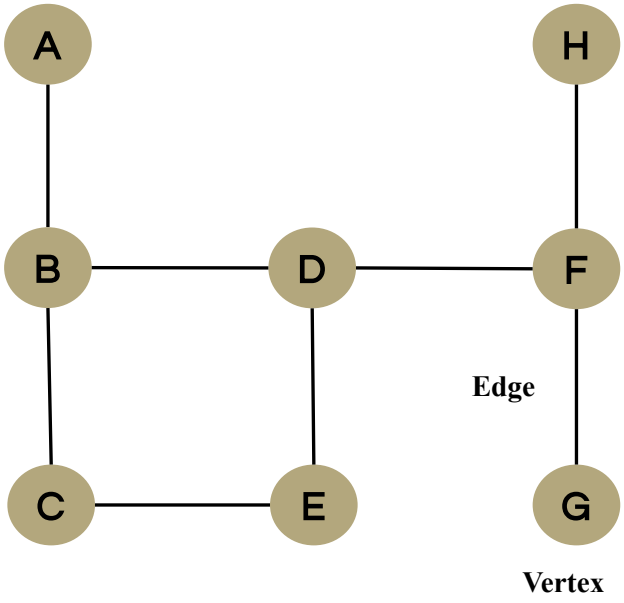


G = (V, E)

V= {A, B, C, D, E, F, G, H}

E= {AB, BC, BD, DE, CE, DF, FG, FH}

Graph



Order of graph-

- Total number of vertices in the graph.

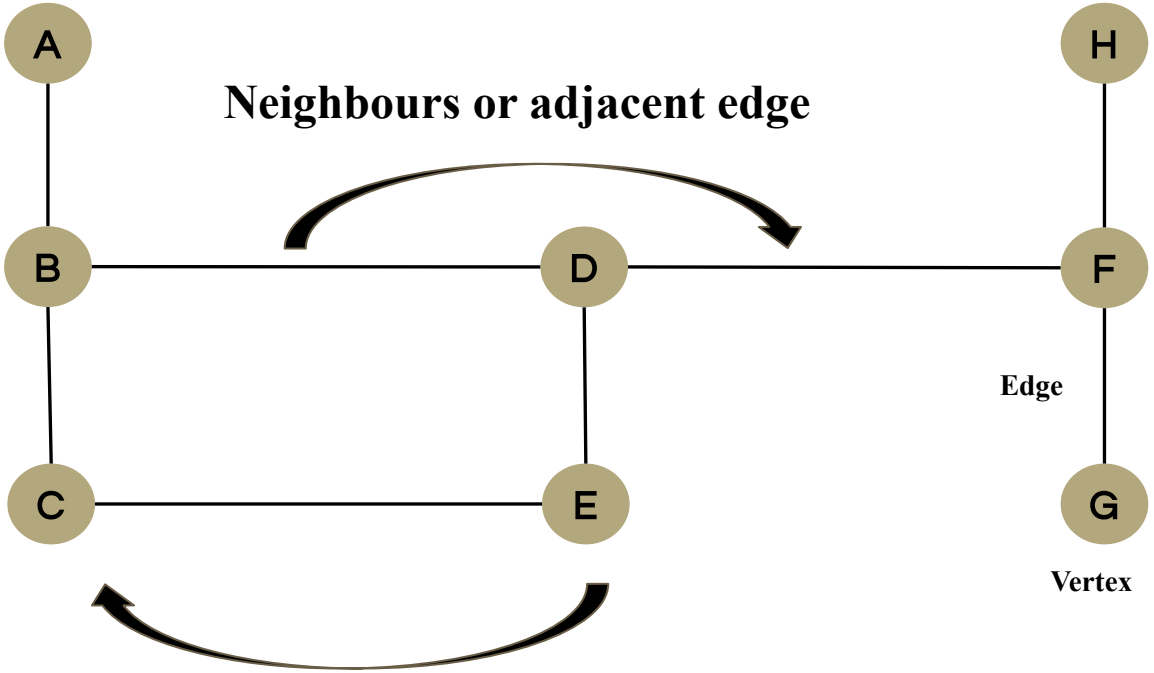
| E | = {AB, BC, BD, DE, CE, DF, FG, FH} = 8

Size of graph-

- Total number of edges in the graph.

| V | = {A, B, C, D, E, F, G, H} = 8

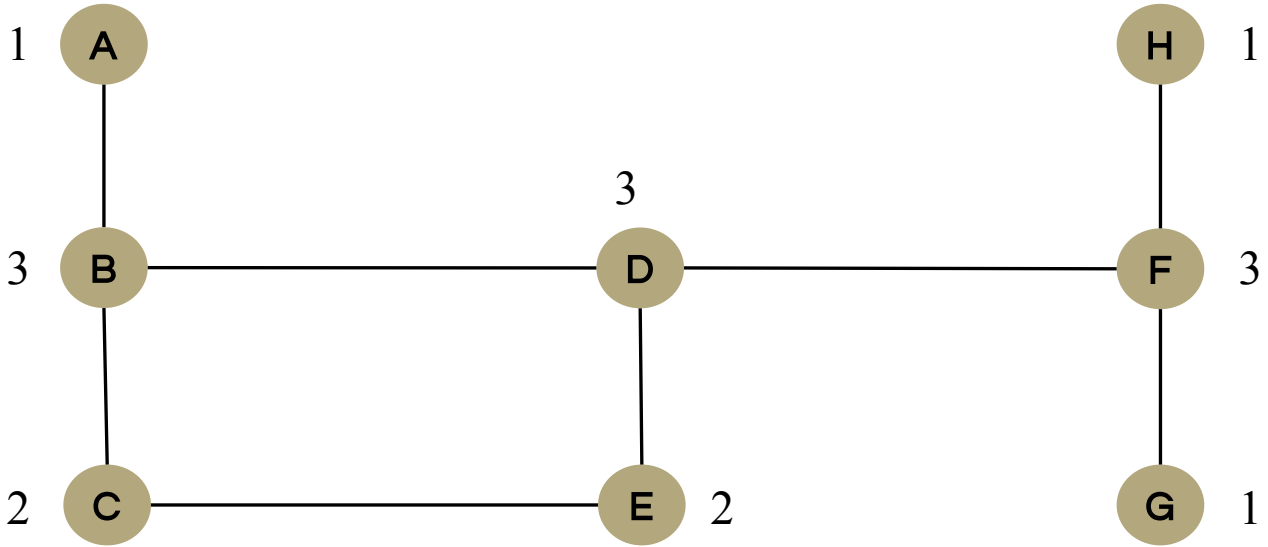
Graph


A
H
Neighbours or adjacent edge
B
D
F
Edge
C
E
G
Vertex
Neighbours or adjacent vertex

Graph

Degree of a vertex-

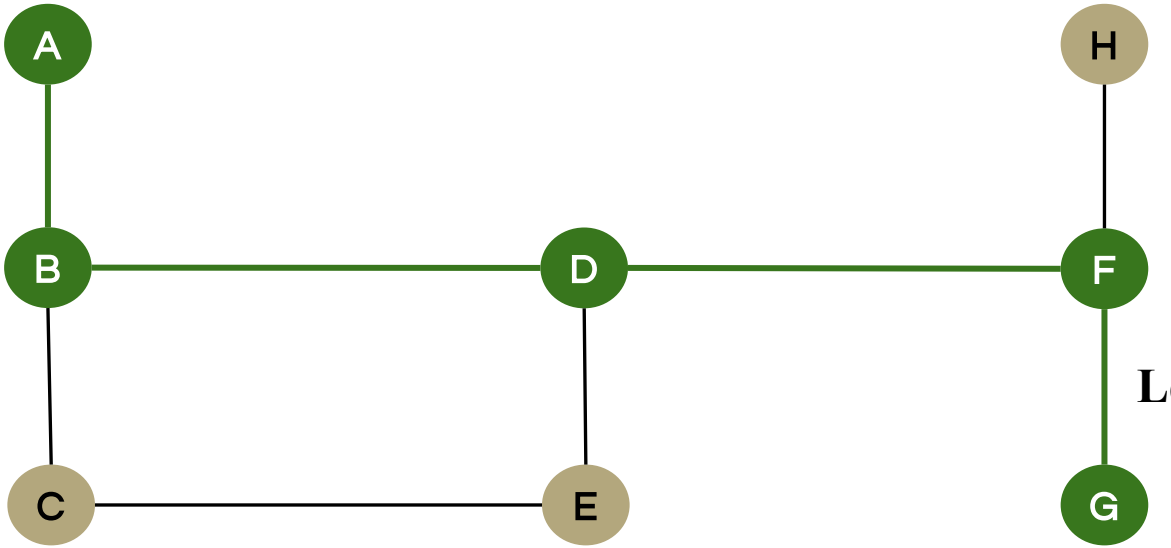
- Number of edges incident with vertex.



Graph

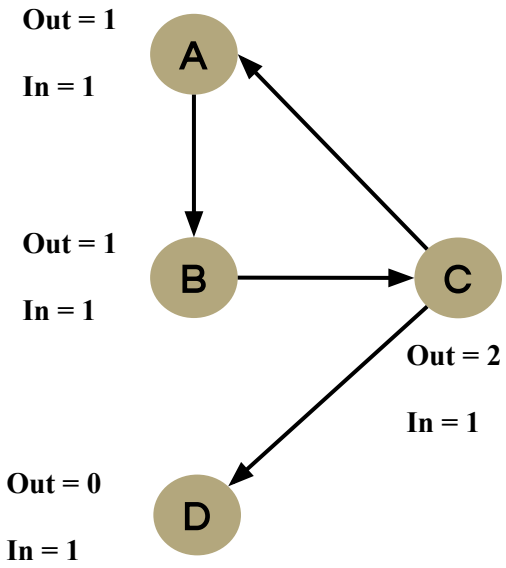
Path-

- Alternating sequence of vertices and edges.
- **Length of path-** Number of edges in path.



Length of path = 4

Directed Graph



Out-degree-

- Number of edges beginning at the node.

In-degree-

- Number of edges ending at the node.

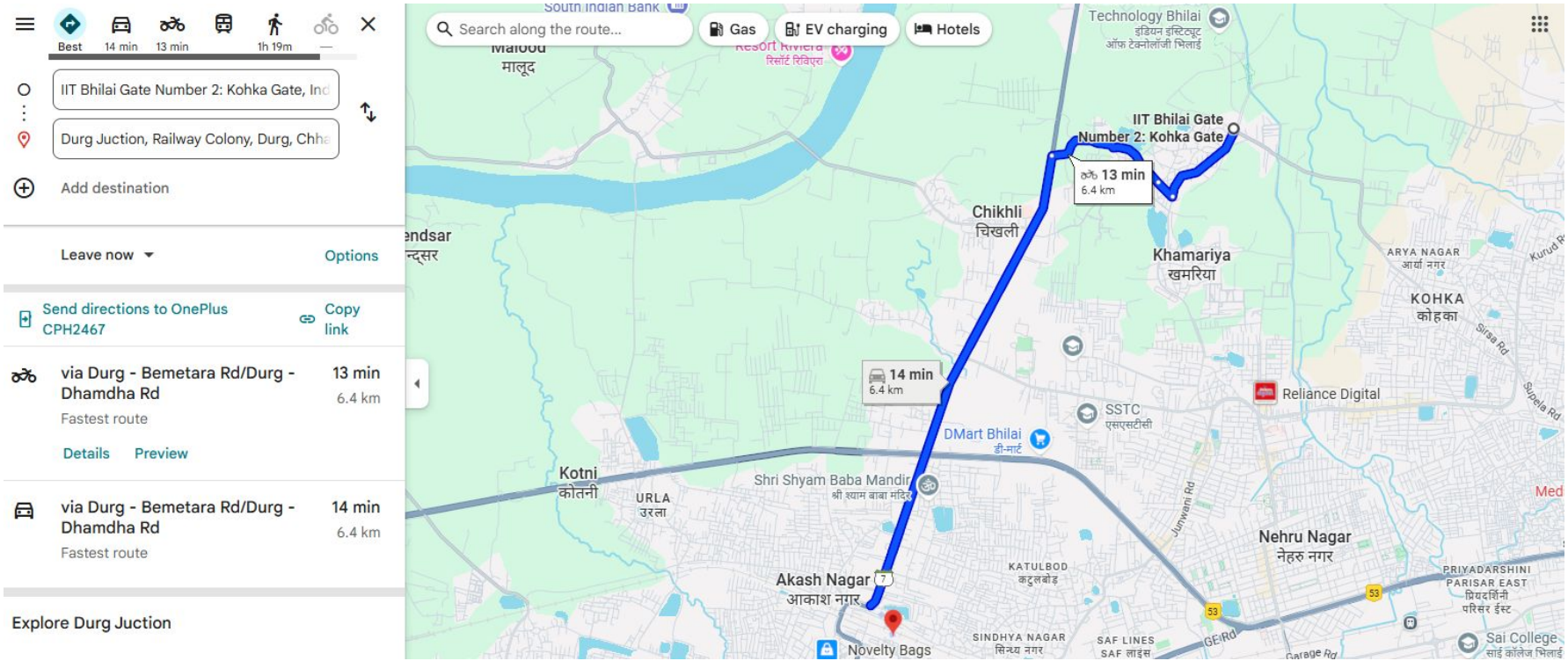


Applications of Graph

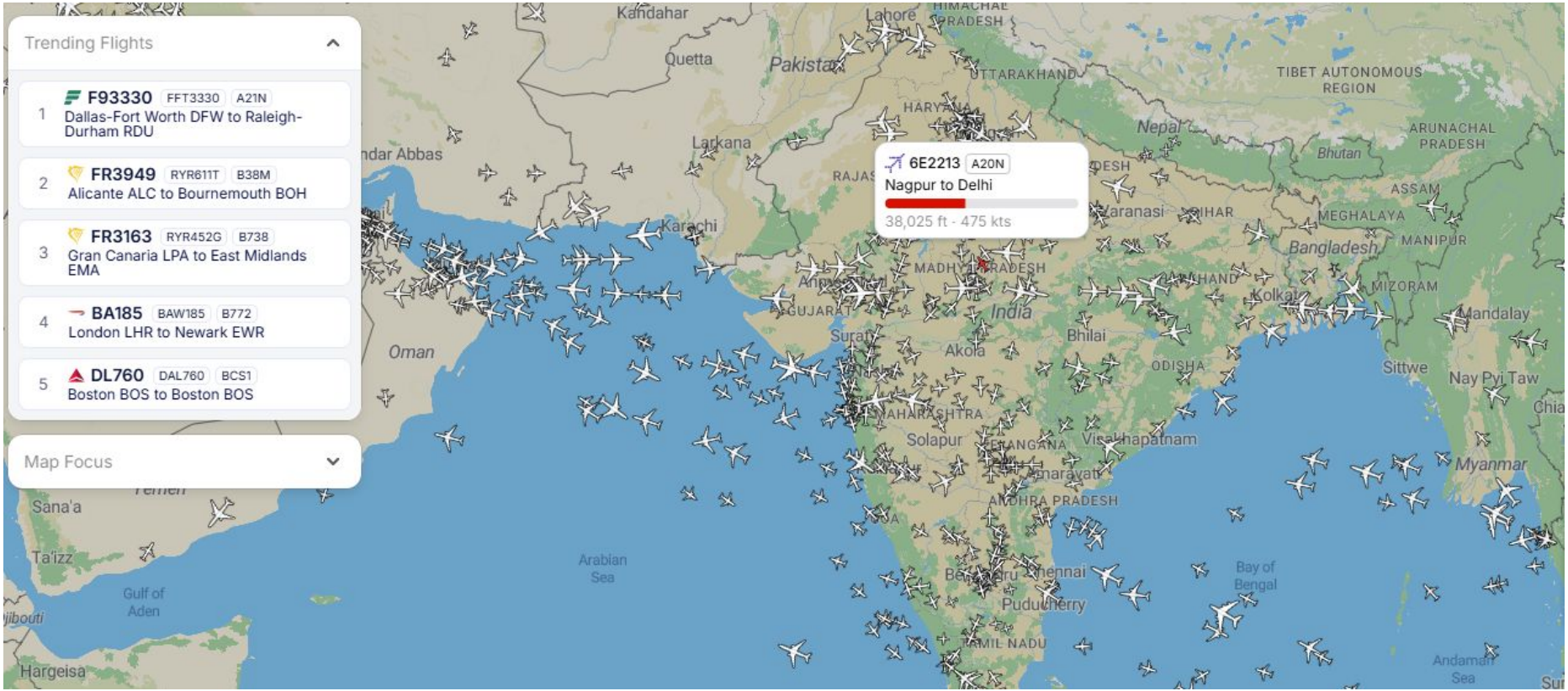
Communication System



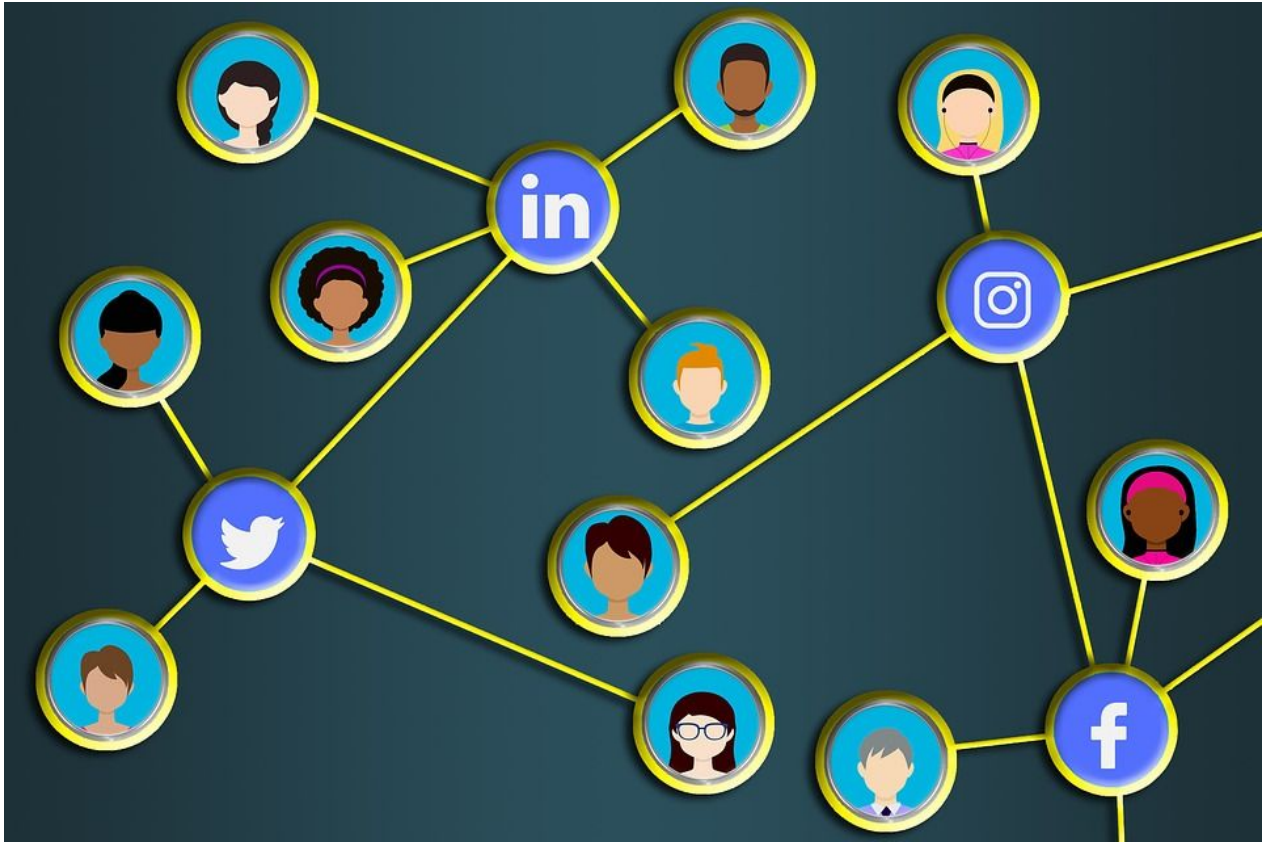
Google Maps


Best
14 min
13 min
1h 19m
IIT Bhilai Gate Number 2: Kohka Gate, Ind
Durg Junction, Railway Colony, Durg, Chha
Add destination
Leave now
Options
Send directions to OnePlus CPH2467
Copy link
via Durg - Bemetara Rd/Durg - Dhamdha Rd
13 min
6.4 km
Fastest route
Details
Preview
via Durg - Bemetara Rd/Durg - Dhamdha Rd
14 min
6.4 km
Fastest route
Explore Durg Junction
Search along the route...
Gas
EV charging
Hotels
Technology Bhilai
IIT Bhilai Gate Number 2: Kohka Gate
13 min
6.4 km
Chikhli
चिखली
Khamariya
खमरिया
ARYA NAGAR
KOHKA
कोहका
14 min
6.4 km
Reliance Digital
SSTC
DMart Bhilai
Kotni
कोतनी
URLA
उरला
Shri Shyam Baba Mandir
Nehru Nagar
नेहरु नगर
Akash Nagar
आकाश नगर
KATULBOD
कटुलबोड
PRIYADARSHINI PARISAR EAST
Novelty Bags
SINDHYA NAGAR
SAF LINES
Sai College

Airlines


Trending Flights
1 F93330 FFT3330 A21N
Dallas-Fort Worth DFW to Raleigh-Durham RDU
2 FR3949 RYR611T B38M
Alicante ALC to Bournemouth BOH
3 FR3163 RYR452G B738
Gran Canaria LPA to East Midlands EMA
4 BA185 BAW185 B772
London LHR to Newark EWR
5 DL760 DAL760 BCS1
Boston BOS to Boston BOS
Map Focus
6E2213 A20N
Nagpur to Delhi
38,025 ft - 475 kts
Kandahar
Quetta
Pakistan
Lahore
Karachi
Larkana
Oman
Yemen
Sana'a
Ta'izz
Gulf of Aden
Hargeisa
Arabian Sea
India
Nepal
Bhutan
Bangladesh
Surat
Akola
Bhilai
Solapur
Visakhapatnam
Amaravati
Bengaluru
Chennai
Puducherry
Bay of Bengal
Andaman Sea
Kolkata
Sittwe
Nay Pyi Taw
Mandalay
Myanmar
TIBET AUTONOMOUS REGION
UTTARAKHAND
HARYANA
ARUNACHAL PRADESH
ASSAM
MEGHALAYA
MANIPUR
MIZORAM
ODISHA
MADHYA PRADESH
MAHARASHTRA
TELANGANA
ANDHRA PRADESH
TAMIL NADU

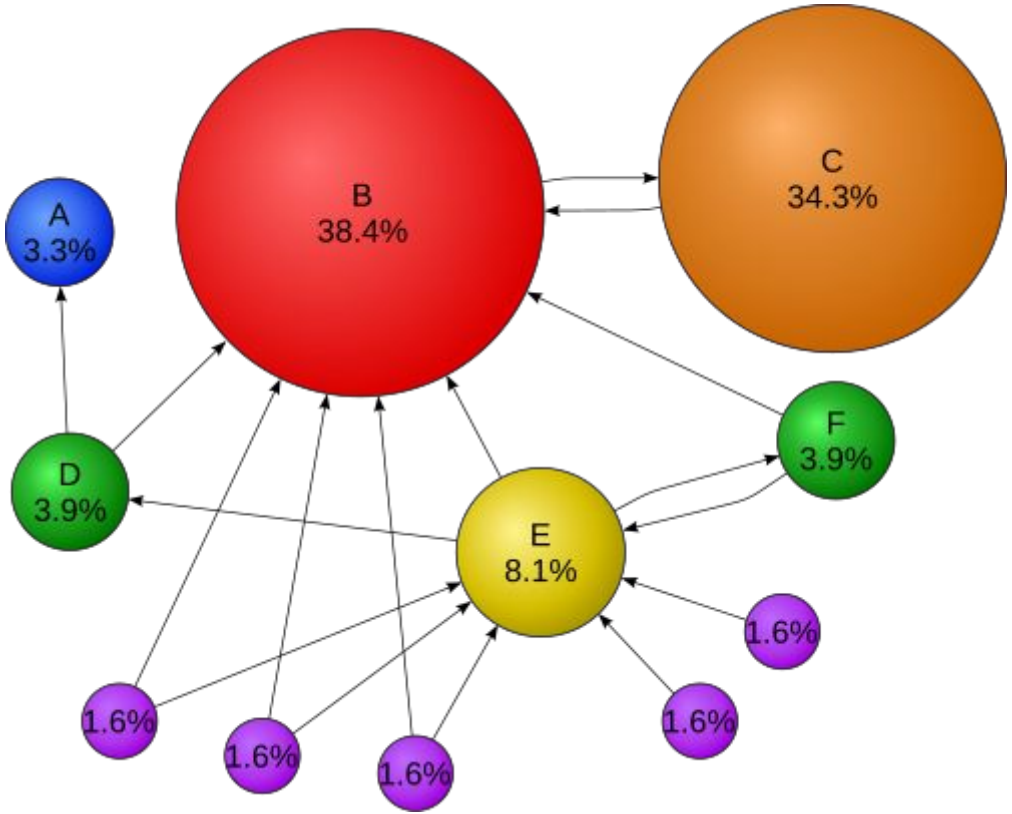
Social Network



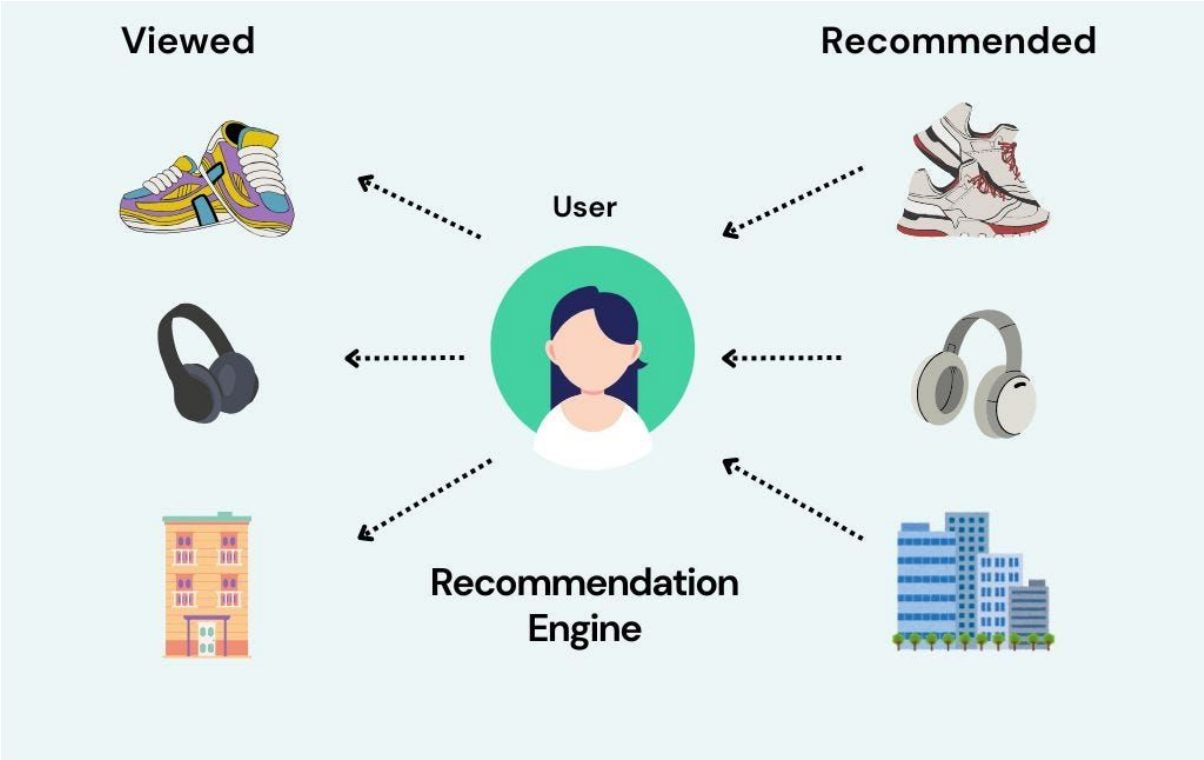
Electrical Network



Web Page Ranking



Recommendation Systems

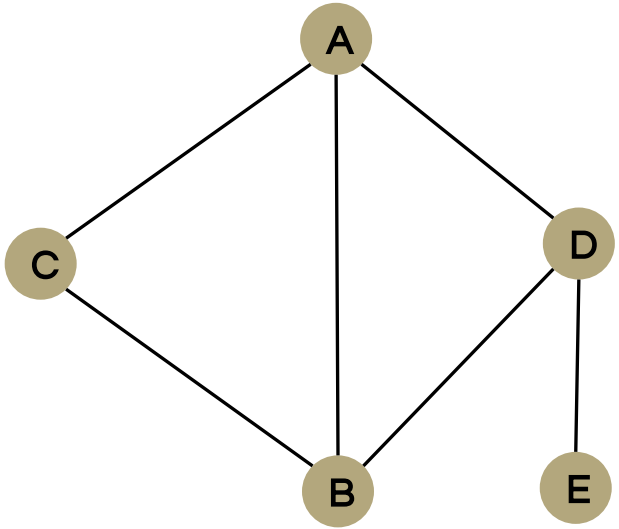




Representation of Graphs

Adjacency matrix Representation

- M[i][j] = 1 → if an edge exists from vertex i to vertex j
- M[i][j] = 0 → if there is no edge

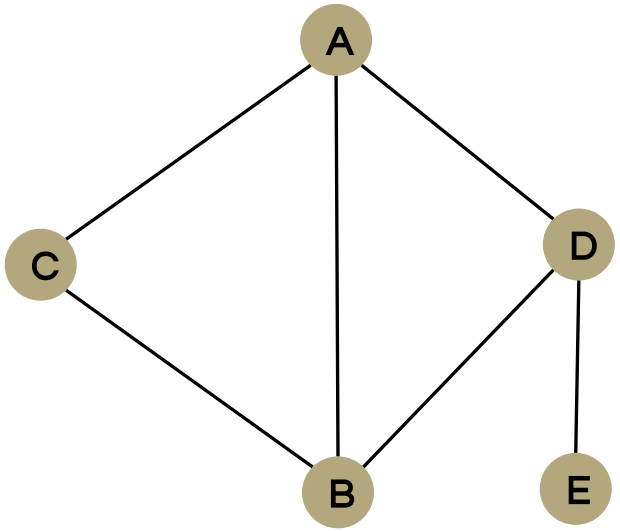


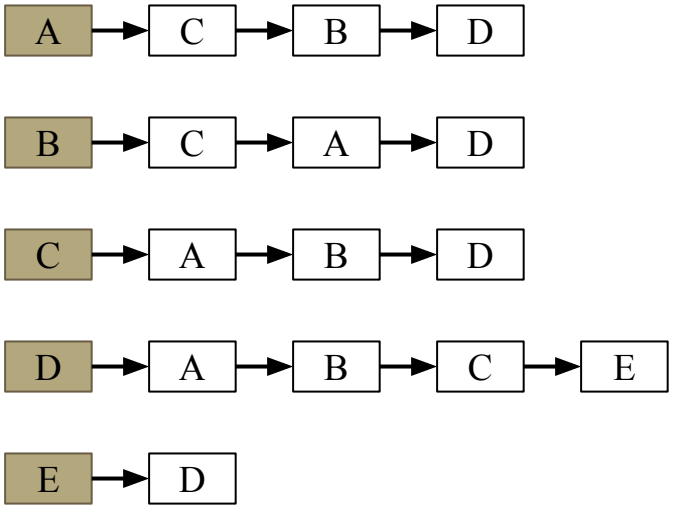
0	1	1	1	0
1	0	1	1	0
1	1	0	0	0
1	1	0	0	1
0	0	0	1	0

- **Space Complexity:** O(|V|²)
- **Time to find all neighbors of vertex i:** O(|V|)
- **Best suited for:** dense graphs (many edges).

Adjacency list Representation

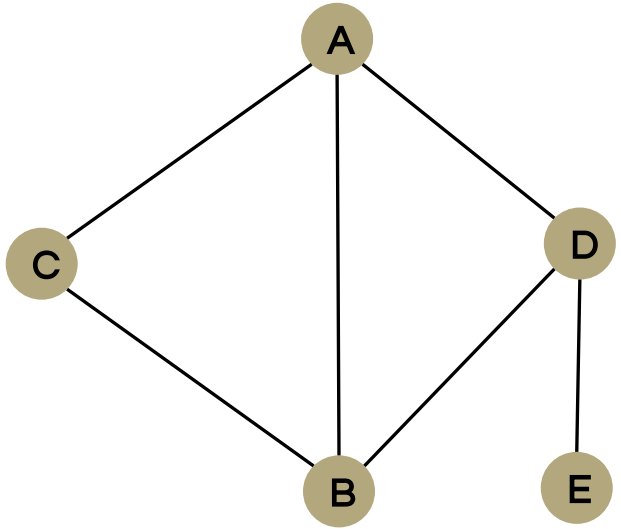
- Each vertex has a linked list containing all the vertices directly connected to it.
- **Space Complexity:** O(V + E)
- **Time to Find All Neighbors of Vertex i:** O(degree(i))
- **Best Suited For:** Sparse graphs.





Edge list Representation

- Each edge is stored as a pair of vertices.
- **Space Complexity:** O(E)
- **Time to Find All Neighbors of Vertex i:** O(E)
- **Best Suited For:** Very sparse graphs.



A	C
A	B
A	D
B	A
B	C
B	D
C	A
C	B
C	D
D	A
D	B
D	C
D	E
E	D

Compressed sparse row (CSR) Representation

- Store sparse graphs using two main arrays:
- **Row Pointer:** Shows where each node's neighbors start.
- **Column Indices:** Stores all neighbors in sequence.
- **Space Complexity:** O(V + E)
- **Time to Find All Neighbors of Vertex i:** O(degree(i))
- **Best Suited For:** Sparse graphs.

